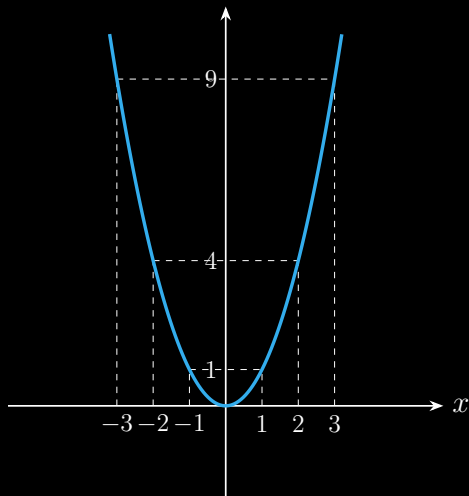
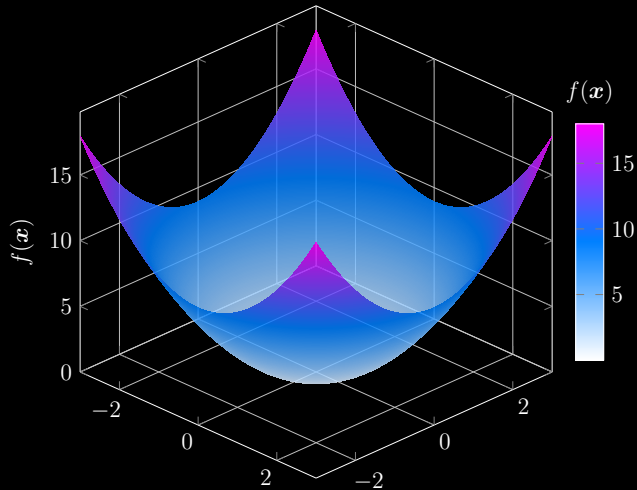


Quadratic Functions $f(x) = ax^2$ vs. $f(x) = x^T Ax$



Positive $a > 0$



Positive definite $A \succ 0$

Positive Definite Matrix $A = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$

Eigenvalue decomposition:

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 5.83 \quad \lambda_2 = 0.17$$

and eigenvectors:

$$\mathbf{u}_1 = \begin{bmatrix} 0.92 \\ 0.38 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.38 \\ 0.92 \end{bmatrix}$$

```
1 import numpy as np
2
3 A = np.array([[5, 2], [2, 1]])
4 eig_val, eig_vec = np.linalg.eig(A)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

Ellipsoid

Representing positive definite matrices as ellipsoids:

$$\mathcal{E}_A = \{\mathbf{x} \mid \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} \leq 1\} \quad (\text{centered at the origin})$$

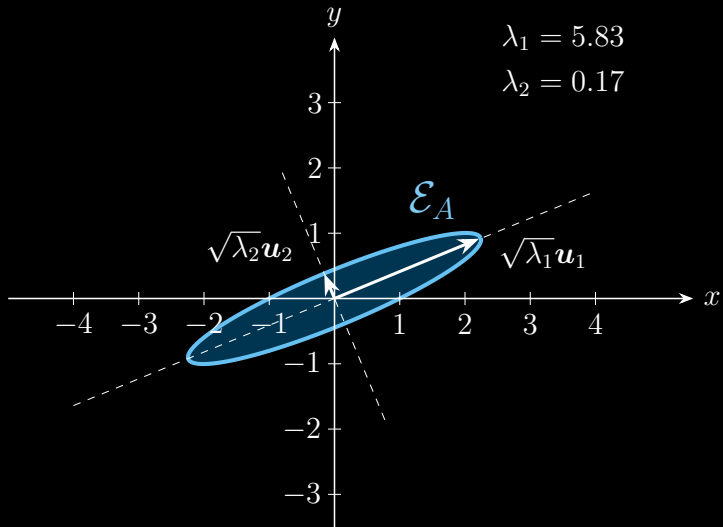
Inverse of eigenvalue decomposition:

$$\begin{aligned} \mathbf{A} &= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top \\ \Rightarrow \mathbf{A}^{-1} &= \frac{\mathbf{u}_1 \mathbf{u}_1^\top}{\lambda_1} + \frac{\mathbf{u}_2 \mathbf{u}_2^\top}{\lambda_2} \end{aligned}$$

Geometric interpretation of $\mathcal{E}_A$:

$$\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} = \frac{(\mathbf{u}_1^\top \mathbf{x})^2}{\lambda_1} + \frac{(\mathbf{u}_2^\top \mathbf{x})^2}{\lambda_2} \leq 1 \quad \rightarrow \quad \text{ellipse!}$$

Ellipse $\mathcal{E}_A = \{ \mathbf{x} \mid \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} \leq 1 \}$



Positive Definite Matrix $B = \begin{bmatrix} 10 & 3 \\ 3 & 2 \end{bmatrix}$

Eigenvalue decomposition:

$$B = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top$$

with eigenvalues:

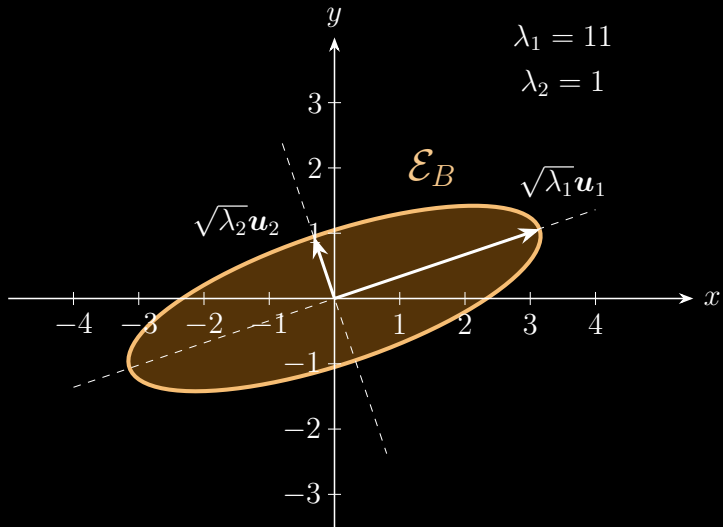
$$\lambda_1 = 11 \quad \lambda_2 = 1$$

and eigenvectors:

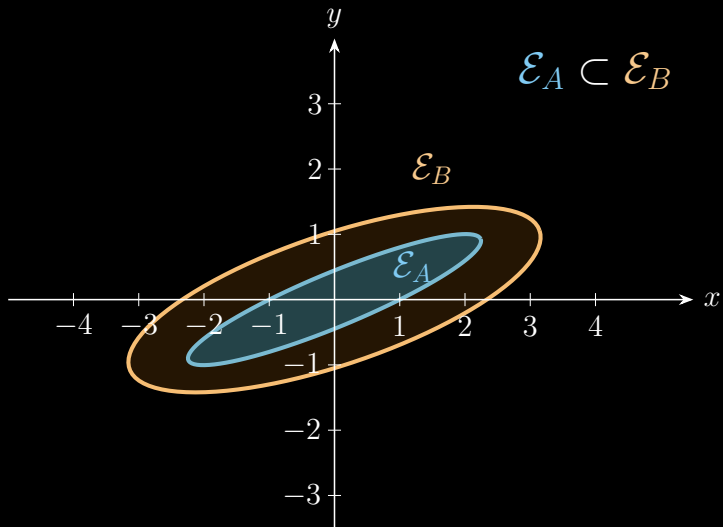
$$\mathbf{u}_1 = \begin{bmatrix} 0.95 \\ 0.32 \end{bmatrix} \quad \mathbf{u}_2 = \begin{bmatrix} -0.32 \\ 0.95 \end{bmatrix}$$

```
1 import numpy as np
2
3 B = np.array([[10, 3], [3, 2]])
4 eig_val, eig_vec = np.linalg.eig(B)
5
6 print('lambda_1 = ', eig_val[0])
7 print('lambda_2 = ', eig_val[1])
8 print('u_1 = ', eig_vec[:, 0])
9 print('u_2 = ', eig_vec[:, 1])
```

Ellipse $\mathcal{E}_B = \{x \mid x^\top B^{-1} x \leq 1\}$



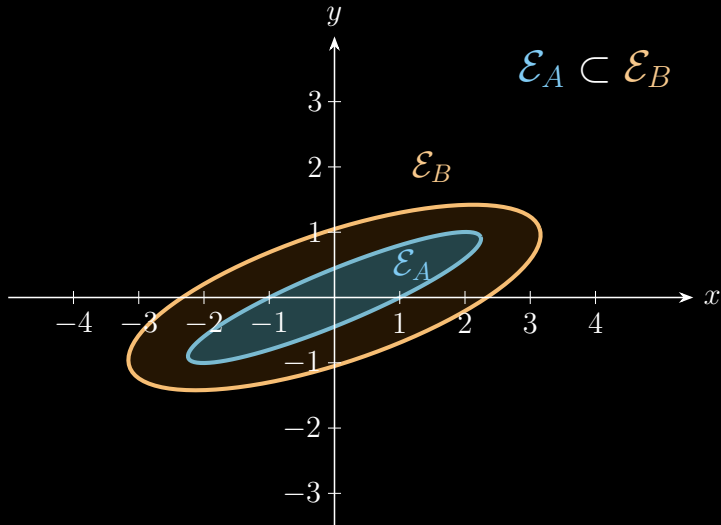
Positive Semidefinite Ordering



Positive Semidefinite Ordering

$$\mathcal{E}_A \subseteq \mathcal{E}_B \quad \text{if and only if} \quad \underbrace{B - A \succeq 0}_{\text{positive semidefinite matrix}} \quad \Leftrightarrow \quad B \succeq A$$

$$B - A = \begin{bmatrix} 10 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 1 \end{bmatrix} \succcurlyeq 0 \quad \Rightarrow \quad \textcolor{red}{B} \succeq \textcolor{red}{A}$$



Matrix Monotone Functions

Thanks for your attention!

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